

Abdul Rahiman Mihad

AI Engineer · Real-Time Conversational AI · Agentic Systems · MLOps · Abu Dhabi, UAE
+971 50 670 8992 | abduhrahimanmihad@gmail.com | [Portfolio](#) | [LinkedIn](#) | [GitHub](#)

PROFESSIONAL SUMMARY

AI Engineer building production real-time conversational AI at HireUp (Japan, remote): a voice-based AI interviewer with sub-second turn latency and a LangGraph-driven interview engine. Previously architected a self-corrective RAG system on AWS Bedrock achieving 92% query accuracy, and an end-to-end YOLOv8 computer vision MLOps pipeline. Expert in LLM reliability engineering, prompt engineering, WebSocket streaming, and cost optimization. Based in Abu Dhabi, available immediately.

CORE COMPETENCIES

Artificial Intelligence (AI) · Machine Learning (ML) · Deep Learning · Generative AI · Large Language Models (LLM) · Natural Language Processing (NLP) · Computer Vision · Conversational AI · Voice AI · AI Agents · Agentic AI · RAG · Prompt Engineering · LLM Fine-Tuning · Model Deployment · MLOps · Data Science · Cloud Computing · Agile

WORK EXPERIENCE

AI/ML Engineer, HireUp May 2026 to Present
Japan HQ, Remote · [app.hireup.now](#) · Built “Alpha”, a production real-time AI Voice Interviewer

- Designed the **full-duplex real-time voice pipeline**: WebSocket streaming with concurrent asyncio loops, Deepgram Flux streaming STT, WebRTC VAD with Groq Whisper fallback, and sentence-level streaming TTS; **cut startup latency by ~1.5s**.
- Achieved 15/15 correct live navigation turns (vs ~80% prompt-only)** by replacing LLM position arithmetic with a deterministic zero-latency NAV resolver; prevented prompt leaks by redacting unseen questions from context.
- Engineered **voice barge-in** with client-side RMS detection in an AudioWorklet, echo rejection, and browser-native AEC cancellation.
- Cut token spend significantly** by replacing vector RAG with one-time JD/CV extraction into compact JSON matrices (~300 to 450 tokens/turn) and zero per-turn summarization calls.
- Built a **text-level eval and regression suite** replaying full interviews in seconds with deterministic checks and LLM judges; wired MLflow metrics and LangSmith tracing.
- Hardened the LangGraph engine with an **11-tag control protocol**, a per-turn intent classifier, behavioral state machines, and a Redis race-condition fix.

PROJECTS

Self-Corrective RAG System LangGraph · AWS Bedrock · FastAPI · MLflow · Docker [Live Demo](#)

- Reduced hallucinations 40%** and reached **92% query accuracy** with a self-corrective multi-agent pipeline: iterative document grading, hallucination detection, and query rewriting using Llama 3.1 (70B) on Bedrock at sub-1.5s latency.
- Exposed agent tools via an **MCP server**; tracked tokens, latency, and cost-per-request in MLflow; deployed on AWS EC2 with FastAPI and Streamlit.

End-to-End Litter Detection System YOLOv8 · CVAT · DVC · MLflow · Streamlit [Live Demo](#)

- Boosted small-object mAP by 15%** on **1,500+ hand-annotated images** using 960px scaling, Mosaic and MixUp augmentation, and automated oversampling for class imbalance.
- Ran a reproducible MLOps workflow with **DVC and MLflow across 50+ experiments**; deployed a real-time Streamlit app with per-class detection counts.

Custom Character-Level Language Model PyTorch · NumPy · Custom Autograd

- Implemented MLPs, backpropagation, and SGD from scratch with a **custom autograd engine**, demonstrating deep understanding of transformer internals.

EDUCATION

Bachelor of Computer Applications (BCA) Aug 2022 to Nov 2025
Yenepoya University, India · Specialisation: ML, AI and Robotics · Industry partnership with TCS and Microsoft

CERTIFICATIONS

- Microsoft Azure**: AI-900 · AZ-900 · DP-900 **Kaggle**: Advanced SQL · Intro to SQL
- IBM / Coursera, Generative AI**: [Advanced LLM Fine-Tuning](#) · [Fine-Tuning Transformers](#) · [AI Agents: RAG + LangChain](#) · [LLM Architecture](#) · [LM with Transformers](#) · [Foundational Models for NLP](#) · [EDA for ML](#)

SKILLS

AI / LLM	LLMs · RAG · Multi-Agent Systems · LangGraph · LangChain · Hugging Face · MCP · PEFT/LoRA · Transformers · NLP · Computer Vision
Real-Time Engineering	WebSockets · asyncio · Deepgram STT/TTS · WebRTC VAD · Web Audio API · Streaming pipelines
MLOps	Python · SQL · JavaScript · PyTorch · TensorFlow · Scikit-learn · XGBoost · FastAPI · REST API · OpenCV
Cloud / Data	MLflow · LangSmith · DVC · Docker · GitHub Actions · CI/CD · Model Deployment · Model Monitoring
	AWS (Bedrock, EC2, S3) · Microsoft Azure · GCP · Redis · PostgreSQL · BigQuery
Languages:	English (Fluent) · Malayalam (Native) · Hindi (Elementary) · Arabic (Elementary)